

**BEFORE THE PUBLIC UTILITIES COMMISSION
OF THE STATE OF CALIFORNIA**



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Order Instituting Rulemaking to
Continue Oversight of Electric
Integrated Resource Planning and
Procurement Processes.

Rulemaking 25-06-019
(Filed June 26, 2025)

**COMMENTS OF THE VEHICLE-GRID INTEGRATION COUNCIL ON ORDER
INSTITUTING RULEMAKING TO CONTINUE OVERSIGHT OF ELECTRIC
INTEGRATED RESOURCE PLANNING AND PROCUREMENT PROCESSES**

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August 1, 2025

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In accordance with the Rules of Practice and Procedure of the California Public Utilities Commission (“Commission”), the Vehicle-Grid Integration Council (“VGIC”) hereby submits these comments on the *Order Instituting Rulemaking to Continue Oversight of Electric Integrated Resource Planning and Procurement Processes* (“OIR”), issued on July 2, 2025.

I. INTRODUCTION

VGIC appreciates the opportunity to submit these comments on the OIR for the next stage of California’s Integrated Resource Planning (“IRP”) proceeding. VGIC appreciates the Commission’s continued efforts to refine the Integrated Resource Planning (“IRP”) framework and ensure that planning and procurement processes are aligned with California’s reliability and decarbonization goals. As the state experiences rapid growth in electric vehicle (“EV”) adoption, VGIC urges the Commission to further incorporate vehicle-grid integration (“VGI”) and EV load flexibility into IRP modeling, scenario development, and procurement planning.

In response to the OIR and the planned development of the 2026 Preferred System Plan (“PSP”), VGIC offers recommendations to improve visibility into VGI-related modeling results, incorporate VGI as a candidate resource, and ensure that bidirectional EV charging potential is

appropriately considered in California’s planning processes. VGIC looks forward to working with the Commission and stakeholders to ensure California’s planning processes fully recognize the evolving capabilities of EVs and the value they can provide to the electric system.

II. BACKGROUND & INTEREST IN PROCEEDING.

VGIC is a 501(c)6 member-based advocacy group committed to advancing the role of EVs and VGI through policy development, education, outreach, and research. VGIC supports the transition to a decarbonized transportation and electric sector by ensuring the value from flexible EV charging and discharging is recognized and compensated to achieve a more reliable, affordable, and efficient electric grid. VGIC’s members include automotive original equipment manufacturers (“OEMs”), EV service providers (“EVSPs”), distributed energy resource (“DER”) aggregators, publicly-owned electric utilities, and Community Choice Aggregators.

Californians are adopting significant numbers of EVs, with over 2 million on the road today. These vehicles can provide significant amounts of energy flexibility for California without compromising customer mobility needs and the movement of goods within and beyond the state. EVs are also unlike any other modifiable load or dispatchable resource type because individual EVs represent a more significant potential load shift and energy storage capacity than the typical customer-sited DERs. VGIC therefore holds a direct interest in shaping grid planning to ensure that the potential for VGI is fully incorporated into California’s planning processes. VGIC has been an active party in related rulemakings on grid planning and transportation electrification such as R.18-12-006, R.21-06-017, R.22-07-005, and R.23-12-008.

III. COMMENTS ON THE PROPOSED SCOPE.

VGIC appreciates the CPUC’s continued efforts to improve the modeling of VGI and demand flexibility from EVs within the IRP. VGIC commends the Commission for considering

the capabilities of EV charging equipment within the IRP modeling effort. In setting the scope and focus of this proceeding, VGIC recommends that the Commission more explicitly consider the potential for EVs to contribute to California’s electric system portfolio in the development of the 2026 Preferred System Plan (“PSP”).

A. The Commission should host a workshop on the results of VGI modeling scenarios for the 2026 PSP.

VGIC was encouraged by the inclusion of VGI in the inputs and assumptions (“I&As”) for the last IRP cycle. However, the results of the VGI-related analysis that was done last cycle were not ultimately published. VGIC is unclear if the sensitivity analyses, including the proposed Mid-VGI and High-VGI scenarios were executed by the modeling team for the 2023 PSP because the results seem to have been withheld.

Since distributed energy resources and EV load flexibility are relatively new within the IRP process, many DER and VGI stakeholders are less familiar with the RESOLVE model. It is difficult to parse through the available RESOLVE spreadsheets to find different results that are not discussed or presented by Energy Division in venues such as Rulings or workshops. To ensure that all stakeholders can understand the results of different sensitivity analyses, VGIC encourages the Commission to include the results of any VGI analyses in published materials and workshops that discuss modeling outputs. A dedicated VGI or load flexibility workshop may also be warranted to ensure there is enough time for discussion on these topics.

Dedicated discussion related to modeling VGI would go a long way in filling a critical knowledge gap for VGI stakeholders, including policymakers and customer groups. VGIC has heard CPUC and CEC Commissioners, as well as senior CAISO representatives, ask for more details on the availability, load/generation curves, and price elasticity of VGI resources on

California's grid. We believe this IRP cycle is a critical opportunity to answer these Commissioner's questions.

B. The Commission should work towards adding VGI as a full candidate resource in the IRP.

VGIC encourages the Commission to incorporate VGI as a full candidate resource in the IRP as soon as possible. There is significant potential for VGI to contribute to California's electric system and provide ratepayer savings. While some modeling has been done on more limited applications of VGI and exploration may soon begin on the ability of VGI to greatly improve distribution system utilization, California has yet to evaluate VGI as a tool in electric system portfolio development. Without explicit consideration of VGI potential in system planning, California risks planning for, and ultimately procuring, excess generation that is not needed due to VGI contributions. This is especially important given the amount of EV charging load that is included in California's demand forecast that is used for the IRP.

Significant work has already been done through the proposed 2026 PSP I&As to gather cost data for VGI resources, develop adoption curves, and create total resource potential estimates. If additional work or data is needed to fully model VGI in the core capacity expansion analysis, the Commission should share what information would allow for full inclusion in the IRP. VGIC understands the complexity of accurately incorporating all aspects of VGI into an IRP model, and we appreciate the inherent challenges of including new resources in general. We would welcome collaboration with Energy Division to continue strengthening VGI representation so that California can evaluate all tools in its toolkit, including the state's fast-growing and flexible customer-owned EV fleet.

C. California can expect significant vehicle export / V2G potential.

The 2025 Draft I&As include vehicle-to-grid (“V2G”) or vehicle export potential as a potential grid resource. VGIC is extremely supportive of modeling V2G as a grid resource and believes that there is significant potential for V2G to contribute to California’s clean energy and reliability needs. California is already home to the largest V2G site in North America located in Oakland providing peak reduction services to Pacific Gas & Electric. California is also unlocking V2G resources in existing programs already. For example, V2G sites may enroll in the Emergency Load Reduction Program (ELRP), the CEC’s Demand Side Grid Support Program (DSGS), PG&E’s V2X Pilots, and SDG&E’s Dynamic Export Rate Pilot. Additionally, SCE has proposed a V2G Export Credit (“Vehicle-to-Grid Resource Proposal”) in its recent General Rate Case Phase II.¹

Automakers are also expected to grow their bidirectional vehicle offerings. More than 50% of OEMs, representing *far greater* than 50% of California EV sales, have publicly stated they will enable vehicles to be V2G capable. Light-duty vehicle auto manufacturers that already offer bidirectional charging capability include: Nissan, Ford, General Motors (including nearly *every* Chevrolet, Cadillac, and GMC EV), Kia, and Tesla. Additional manufacturers have made public statements clarifying their intent to offer V2G solutions, including Hyundai, Volkswagen, BMW, Mercedes-Benz, Lucid, Audi, Volvo, Polestar, and Rivian. Medium and heavy-duty vehicles, especially school buses, are also expected to have significant V2G potential. World Resources Institute (“WRI”) notes that California has over 3,110 committed electric buses across the state, of which nearly 45% are delivered or operating.² A majority of these electric school buses will be

¹ See A.24-03-019.

² Lydia Freehafer, Leah Lazer, and Brian Zepka. *The State of Electric School Bus Adoption in the US*. Updated November 18, 2024. <https://www.wri.org/insights/where-electric-school-buses-us>.

Type C, which come equipped with between 126 kWh and 315 kWh of battery capacity.³ Moreover, *all* school buses claiming credits under California Air Resources Board’s Hybrid and Zero-Emission Truck and Bus Voucher Incentive Project (“HVIP”) *or* Zero-Emission School Bus and Infrastructure (“ZESBI”) programs, which represent the vast majority of California’s electric school buses, are **currently required to be capable of bidirectional charging**.

Given the significant potential, VGIC encourages the IRP team to conduct sensitivity analyses with higher amounts of V2G to ensure that this resource can be optimized alongside California’s electric system portfolio.

IV. PROCEDURAL SCHEDULE, CATEGORIZATION, AND NEED FOR HEARINGS.

VGIC supports the categorization and proposed schedule of this proceeding and agrees with the preliminary determination that hearings may be needed.

V. NOTICES.

Service of all notices and communications in this proceeding should be directed to the following VGIC representative:

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³Jessica Wang, Vishant Kothari, Emmett Werthmann, Eleanor Jackson, Lidia Henderson, Alissa Huntington. Updated August 2024. “Electric School Bus Market Study and Electric School Bus U.S. Buyer’s Guide” Washington, DC: World Resources Institute. <https://electricschoolbusinitiative.org/electric-school-bus-market-study-and-electric-school-bus-us-buyers-guide>

VI. CONCLUSION.

VGIC appreciates the opportunity to provide these comments on the OIR. We look forward to further collaboration with the Commission and stakeholders on this initiative.

Respectfully submitted,

/s/ Zach Woogen

Zach Woogen

Executive Director

VEHICLE-GRID INTEGRATION COUNCIL

Date: August 1, 2025